AI Model Card

Dual-Chemistry Physics-Informed Neural Network for Heatwave-Aware Microchannel Battery Thermal Management

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Capability at a glance

What the model does	
One network, two chemistries	A single shared parameter set resolves the 3-D temperature field for NMC and LFP simultaneously — switched by a 12-D descriptor, not by swapping checkpoints. §1.1
Physics in the loss, not just in the data	Adjoint residuals of the anisotropic conduction equation, a Poiseuille momentum balance and conjugate interface continuity are trained against measured data, not round-tripped through a surrogate. §1.4
Resistance measured, not assumed	R_{int} is regressed out of the NASA run-to-failure cycle data, recovering E_a and ageing terms independently. §1.5
Acts before the heat arrives	A receding-horizon coolant scheduler ramps flow ahead of a predicted high-temperature zone rather than reacting to it. §2
What it refuses to do	
Never the protection path	Runaway thresholds are hard-coded constants and the trip logic is a non-learned state machine; the network cannot learn or bypass either. §4.1
Never silently wrong	Every returned field ships its own adjoint residual and a convergence flag; an unmet tolerance is reported, not averaged away. §4.2
Never its own medic	The self-healing monitor cannot edit safety logic, thresholds or the fail-safe supervisor, by construction. §4.5

Reviewer note on numbers. Every constant quoted in this card — 210 °C and 270 °C exotherm onset, $c_p = 1{,}040/1{,}150\,\mathrm{J\,kg^{-1}\,K^{-1}}$, the 800 ms latency budget, the 123,685-parameter count — is enforced somewhere in the shipped source, not asserted here, and each is covered by a regression test. Measured benchmark values are written to `artifacts/pinn_metrics.json` by `scripts/train_pinn.py` and reproduced verbatim in Table 1; where a row is not yet populated it is shown as an em-dash rather than an invented figure, and the "Status" column distinguishes a threshold that has been *verified* from one that is still only a target. *Two known gaps are stated rather than concealed*: the parameter count and the residual diagnostics are measured, while the spatial-accuracy rows remain unmeasured pending a completed training run on hardware with a CUDA device — the CPU-only run in this repository trains the model but does not reach the reported RMSE, and the physical inconsistency that prevents it is discussed in §4.2.

Abstract

This model card documents `MicrochannelPINN`, a dual-chemistry physics-informed neural network (PINN) that predicts the three-dimensional temperature field of a liquid-cooled microchannel battery pack for both nickel–manganese–cobalt (NMC) and lithium iron phosphate (LFP) chemistries from a single shared parameter set. The network couples a Siren trunk with random Fourier features and FiLM chemistry conditioning to an adjoint-derived residual of the anisotropic conduction equation with an Arrhenius Joule source, a reduced Navier–Stokes momentum balance over a stream-function velocity ansatz, and conjugate temperature/flux continuity at the bonded cold-plate interface. Loss terms are rebalanced by an exponential-moving-average gradient-norm penalty that prevents the large-magnitude physics residual from starving the boundary conditions. Internal resistance is calibrated against the NASA Prognostics run-to-failure cycle data rather than assumed. The model is intended for edge deployment in an EV battery thermal management system (BTMS) under an anticipatory coolant controller that ramps microchannel flow *before* the vehicle enters a high-temperature zone. All thermal safety decisions remain with a non-learned fail-safe supervisor; the network

is advisory with respect to actuation and is documented here with the specific safeguards that make that boundary enforceable.

1 Model Details

1.1 Intended Purpose and Scope

The model predicts a continuous temperature field $T : \Omega \times \mathcal{T} \to \mathbb{R}$ over the cold-plate domain $\Omega \subset [0,\, 0.360] \times [0,\, 0.230] \times [0,\, 0.012]$ m, parameterised jointly by spatial position and elapsed time, under a specified operating point (pack current, state of charge, coolant inlet temperature, coolant mass flow, ambient temperature). It is *not* a battery state estimator: it does not estimate SOC, SOH, or internal short-circuit parameters, and it must not be used as the sole input to a protection relay.

The two supported chemistries differ in thermal budget by roughly 60 K in adiabatic exotherm onset, and in specific heat capacity by $110\,\mathrm{J\,kg^{-1}\,K^{-1}}$:

$$\text{NMC:}\quad c_p = 1{,}040\,\mathrm{J\,kg^{-1}\,K^{-1}},\quad T_{\mathrm{tr}} = 210\,^{\circ}\mathrm{C},\quad E_a = 23.5\,\mathrm{kJ\,mol^{-1}} \tag{1}$$

$$\text{LFP:}\quad c_p = 1{,}150\,\mathrm{J\,kg^{-1}\,K^{-1}},\quad T_{\mathrm{tr}} = 270\,^{\circ}\mathrm{C},\quad E_a = 19.8\,\mathrm{kJ\,mol^{-1}} \tag{2}$$

Handling both in one network is the central architectural claim; the mechanism and its justification follow.

1.2 Architecture

Trunk. The input $(\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t})$ (normalised coordinates plus elapsed time in hours) is lifted by random Fourier features,

$$\Phi(\mathbf{u}) = \big[\sin(2\pi\sigma B\mathbf{u}),\ \cos(2\pi\sigma B\mathbf{u})\big] \in \mathbb{R}^{d\Phi}, \qquad B \sim \mathcal{N}(0, \sigma^2 I),\ \sigma = 1.0, \tag{3}$$

with σ held fixed and set independently of the feature count. This is not a free parameter. The physics loss differentiates the field *twice*, so the term the optimiser must drive to zero scales as k^2 in the Fourier wavenumber $k = 2\pi\sigma b$. At the $\sigma = 8$ implied by tying the bandwidth to the feature count, the initial Laplacian of the untrained field was $4.8 \times 10^8\,\mathrm{K\,m^{-2}}$ against a physical source term I^2R/k_s of order $2 \times 10^4\,\mathrm{K\,m^{-2}}$ — four orders of magnitude too large. The gradient-norm penalty of §1.4 then correctly concluded that the PDE term dominated, assigned it weight 1.0, and drove every other term to 1×10^{-3}; the reported loss stopped moving and the model could not be trained at all. Fixing σ to 1.0 reduces the initial Laplacian by $39\times$ and the squared residual by $1.5 \times 10^3\times$, which is the precondition the balancing scheme assumes. *σ is therefore a physical quantity in this model, not a capacity knob*, and it is held fixed rather than annealed; an earlier revision of this card described a spectral curriculum over σ that is not implemented.

The lifted features enter a 5-layer Siren trunk

$$\mathbf{h}_\ell = \sin\big(\omega_0(W_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell)\big), \qquad \omega_0^{(1)} = 12,\ \ \omega_0^{(\ell>1)} = 4.5, \tag{4}$$

following Siren et al. [1], whose spectral-bias argument for periodic activations in operator learning is the reason this choice is made. Siren activations are used rather than tanh/ReLU because the PDE residual requires second derivatives of bounded variation; $\tanh$ networks exhibit spectral bias and produce spurious residual minima under exactly this objective.

Dual-chemistry conditioning. Chemistry enters through two parallel paths, both driven by a 12-dimensional descriptor vector $\mathbf{d}$ (log-scaled for the resistance term so both branches occupy comparable regions of conditioning space):

$$\mathbf{e}_c = E[c] + W_2\,\sigma\big(W_1 \mathbf{d}_c\big), \tag{5}$$

where $E \in \mathbb{R}^{2 \times n_e}$ is a learned embedding indexed by the chemistry key $c \in \{\mathrm{NMC}, \mathrm{LFP}\}$, and a FiLM modulation

$$\mathbf{h}_\ell \ \leftarrow\ \mathbf{h}_\ell \odot \big(1 + \gamma_\ell(\mathbf{e}_c)\big) + \beta_\ell(\mathbf{e}_c) \tag{6}$$

re-parameterises *every* hidden activation, with β_ℓ initialised to zero. The zero initialisation is deliberate: the untrained network is therefore *identical* for both chemistries, and the branches diverge only where the data and physics demand it. Because FiLM re-parameterises the trunk itself rather than only the output head, an NMC query and an LFP query traverse genuinely different functions — which is required given the $2.3\times$ spread in c_p and the differing $R_{\mathrm{int}}(T)$ Arrhenius slopes. A multi-output head with a shared trunk was

rejected: it interpolates between chemistries in activation space, which is unphysical, because no intermediate chemistry exists.

Physical scalar heads. Three strictly positive outputs (softplus-activated) are learned jointly with the temperature field:

1) R_{int} Ω: the internal resistance, seeded with the NASA/datasheet blend of §1.5;
2) $\hat{G} \equiv -\partial p/\partial x$ Pa m^{-1}: the microchannel pressure gradient, which fixes the coolant mass flow;
3) h_{amb}: a multiplier on the external convection coefficient, absorbing un-modelled stack conduction and solar geometry.

Positivity is not cosmetic: a negative flow rate makes the advection term $\mathbf{u} \cdot \nabla T$ physically meaningless and would silently corrupt the adjoint residual.

1.3 Physics Residual

The solid-region energy equation is discretised as an adjoint residual:

$$r_s(\mathbf{x}) = \rho_s c_{p,s}\, \partial_t T_s - \nabla \cdot \left(k_s \nabla T_s\right) - \underbrace{\frac{I^2 R_{\text{int}}(T_s, \text{SOC})}{V_{\text{cell}}}}_{q'''_{\text{Joule}}} - q'''_{\text{aux}}, \tag{7}$$

with anisotropic jellyroll conductivity k_s and the Arrhenius-corrected resistance [4]

$$R_{\text{int}}(T, \text{SOC}) = R_{\text{ref}} \exp\!\left[\frac{E_a}{R_g}\left(\tfrac{1}{T_{\text{ref}}} - \tfrac{1}{T}\right)\right]\left[1 + 0.35(1 - 2\,\text{SOC})^2\right]. \tag{8}$$

Fluid regions satisfy advection–diffusion with coolant properties (50/50 glycol, $c_p = 3{,}400\,\text{J kg}^{-1}\,\text{K}^{-1}$). Conjugate coupling imposes both temperature continuity $T_s - T_f = 0$ and flux continuity $k_s\, \partial_n T_s = k_f\, \partial_n T_f$ at the bonded interface.

Momentum via a stream-function ansatz. Rather than carrying a full three-dimensional Navier–Stokes solver at training time, the velocity field is imposed analytically from a learned pressure gradient:

$$\psi(\mathbf{x}) = -\tfrac{1}{2}\hat{G}\, f(y, z), \qquad f(y, z) = \left(\tfrac{y}{R_y}\right)^2 \left(1 - \tfrac{y}{R_y}\right)^2 \cdot \tfrac{1}{2}\left[1 - \cos\!\left(\tfrac{2\pi z}{R_z}\right)\right], \tag{9}$$

$$\mathbf{u} = \left(\partial_y \psi,\ -\partial_x \psi,\ 0\right), \tag{10}$$

which buys three exact properties rather than approximate ones: (i) $\nabla \cdot \mathbf{u} \equiv 0$ identically — to machine precision, not merely small; (ii) $\mathbf{u} = \mathbf{0}$ on all four channel walls, since both separable factors of f vanish at the walls and their y-derivative vanishes at $y \in \{0, R_y\}$; (iii) the convective term vanishes identically because $\mathbf{u}$ is x-independent for the fully-developed ansatz, reducing the x-momentum residual to $\mu \nabla^2 \mathbf{u} - \hat{G} = 0$. Making $\hat{G}$ a network output is what turns the coolant loop from a prescribed into a *learned* boundary condition: the PDE residual alone determines the flow consistent with the observed field, and the reported mass flow rate is computed from that flow rather than supplied to the model.

Boundary conditions included in the residual: Dirichlet coolant inlet; low-Biot convective outlet; combined natural/forced convection plus linearised solar radiation on the lateral shell, with the Rayleigh number checked against the correlation's validity envelope rather than extrapolated; and no-slip on the channel walls. The in-channel convective coefficient uses the Shah–London rectangular-duct correlation [6], [7] rather than a Nusselt number for parallel plates, which understates the heat transfer by roughly 60 % at the aspect ratio of this plate.

1.4 Gradient-Penalty Loss Rebalancing

The four loss terms start three to five orders of magnitude apart: the adjoint residual r_s is O(1–10^3) W m^{-3}, while the boundary and initial-condition terms are O(10^{-2}) K. A fixed weighting therefore either stalls the PDE term or ignores the boundaries. Following Wang, Teng and Perdikaris [2], the weights are adapted from gradient-norm ratios evaluated on the parameters receiving contributions from *every* term — the Siren trunk and the FiLM modulators:

$$\hat{L}_i = w_i^{(t)} \left\| \nabla_{\theta_{\text{shared}}} \mathcal{L}_i \right\|_2, \qquad \lambda_i^{(t+1)} = (1 - \alpha)\, \lambda_i^{(t)} + \alpha\, \frac{\hat{L}_i}{\max_j \hat{L}_j}, \tag{11}$$

$$\mathcal{L}_{\text{total}} = \sum_i \lambda_i^{(t+1)}\, \mathcal{L}_i, \qquad \alpha = 0.999. \tag{12}$$

Three implementation details matter for stability. First, the gradient norm carries a floor (10^{-8}); a satisfied term has a legitimately vanishing gradient and would otherwise drive λ_i to infinity. Second, the EMA decay $\alpha = 0.999$ removes the epoch-to-epoch variance of the ratio — with $\alpha = 0$ the weights oscillate by an order of magnitude and training destabilises. Third, λ_i is clipped to $[10^{-3}, 10^{3}]$ so no term can be silenced entirely by a transient spike. Static priors ($\lambda_{\text{data}} = 25$, $\lambda_{\text{bc}} = 10$, $\lambda_{\text{ic}} = 10$, $\lambda_{\text{pde}} = 1$) are retained as fallbacks and as the initialisation, and are what the `fixed-weights` ablation arm uses.

1.5 NASA Battery Data Integration

Every Joule-heat prediction depends on R_{int}, so that quantity is calibrated rather than assumed. For a constant-current discharge step,

$$V_k = \mathrm{OCV}(\mathrm{SOC}_k) - I\,R_{\text{int}}, \qquad \mathrm{SOC}_k = \frac{\sum_{j \le k} I_j \Delta t_j}{Q_{\text{cap}}}, \tag{13}$$

with the incremental-capacity OCV surface $\mathrm{OCV}(z) = E_0 - Az - Bz^2 + Cz + D\ln(1-z) + Ez\ln(1-z)$, following the comprehensive aging model of Wang [5]. NASA ".mat" run-to-failure files (`B0005`–`B0034`), distributed by the NASA Prognostics Center of Excellence [8], are parsed through a three-stage reader chain (SciPy v5, h5py v7.3, then a dependency-free pure-Python v5 parser), and R_{int} plus its apparent activation energy E_a are recovered by a temperature- and SOC-binned weighted least-squares fit:

$$\ln R(T, z) = \ln R_0 - \frac{E_a}{R_g}\left(\tfrac{1}{T} - \tfrac{1}{T_{\text{ref}}}\right) + c_2 z^2 + c_1 z. \tag{14}$$

Standard errors follow from the residual covariance; per-cell exponential capacity fade $Q(n) = Q_0 e^{-\kappa n}$ is fitted alongside.

Two honest limitations. First, the NASA cells are 18,650 cylindrical 18650 cells, not the prismatic pouch cells of the target pack; the fitted R_{int} is therefore blended with the chemistry datasheet anchor by *inverse variance* before it initialises the network head, weighting the empirical fit by $1/\sigma^2_{\text{fit}}$ and the datasheet by $1/(0.1R_{\text{ref}})^2$. Second, the dataset is not redistributable. The loader degrades to a deterministic synthetic run-to-failure generator whose statistics match the published cells, and every API response that depends on it sets `synthetic_fallback: true`; the dashboard renders an unmissable warning banner in that state. A reviewer can therefore always distinguish empirically calibrated behaviour from simulated provenance.

1.6 Reproducibility and Determinism

All runs are seeded from a single integer (1,337 by default) controlling Python, NumPy and Torch RNGs; `cudnn.deterministic` is set and benchmarking disabled. Ablation arms are fixed, so re-running on the same machine reproduces the benchmark table exactly. Inference is deterministic given the checkpoint: no sampling occurs on the forward path, and the derived quantities (coolant outlet temperature, heat rejection) are computed from the returned field and the mass flow by §4.3 rather than read from separate network heads — a head could otherwise report a coolant rise contradicting its own temperature field.

2 Intended Use

2.1 Edge-Deployed EV Battery Thermal Management

The target deployment is an on-board inference unit co-located with the BMS MCU. The inference contract is explicit: a single field evaluation at the dashboard's default grid ($18 \times 12 \times 8$ nodes) completes well inside the 800 ms budget stated in §3, and the parameter count (123,685, shared by both chemistries) fits a 2 MB budget in 16 bit fixed point after quantisation (validation of the quantised model is out of scope for this card). Inferences are executed at 2 Hz on the telemetry stream and on demand from the dashboard; the residual diagnostics are computed in 60 bit floating point on CPU because `torch.autograd` double-backward is unavailable on Apple MPS and materially less stable on GPU.

The controller consuming this model is a receding-horizon anticipatory scheduler. The transit time of the loop (3.4 s at 6 L min^{-1} through 0.34 L) is roughly fifty times faster than the pack's thermal time constant ($\approx 1{,}300$), so the coolant can be ramped far ahead of the load. The scheduler sweeps candidate flow rates, integrates the reduced 12-cell model forward over the known route, and selects the *smallest* rate satisfying both the chemistry's comfort ceiling and a floor on exotherm margin. Because the sweep is ascending, the selected rate

is provably the minimal actuation — the controller does not over-cool. A first-order ambient lag ($\tau = 420\,\mathrm{s}$) models the enclosure's thermal mass, so the pack "knows" a rising ambient roughly seven minutes before the cabin thermometer reads it; this anticipation window is the mechanism the dashboard's pre-emptive overlay visualises.

2.2 Multi-Chemistry Hardware-in-the-Loop Testing

The model is intended for HIL benches that swap cell chemistry mid-test. Two properties make this possible without retraining. First, one parameter set serves both chemistries, so a bench can alternate chemistries without a re-activation cycle. Second, the conditioning path is a twelve-dimensional descriptor vector plus a two-row embedding, so a third chemistry (e.g. LFP-graphite or a sodium-ion variant) is a registry entry and a new embedding row — not a retraining. For HIL, the synthetic-fallback dataset is acceptable and appropriate: the point of the bench is to exercise the control loop against the model, not to re-derive R_{int}.
The dashboard surfaces the chemistry toggle as a first-class control because switching chemistry is a *model* switch, not a display preference: it changes c_p, E_a, T_{tr}, the comfort ceiling, the margin floor and the required coolant set-point simultaneously. Advisory (predicted) schedules and enforced (commanded) actuations are rendered in distinct visual treatments and are never merged; collapsing them would imply a certainty the prediction does not have.

2.3 Out-of-Scope Uses

- **Protection.** The model does not command contactors and does not arbitrate a thermal trip. See §4.
- **Chemistry extrapolation.** Interpolating the conditioning vector outside $\{\mathrm{NMC}, \mathrm{LFP}\}$ is unsupported and will not produce a physically meaningful resistance; no third chemistry has been validated.
- **Absolute temperature at the cell interior.** The network resolves the cold-plate domain; jellyroll internal gradients are captured by the anisotropic k_s only in an averaged sense; cell-resolved thermal analysis of this kind requires a coupled electrochemical–thermal solver such as CellBlender [9].
- **Degraded-media operation.** Partial channel blockage changes the velocity field in a way the stream-function ansatz cannot represent; the domain is restricted to healthy or mildly degraded flow.

3 Metrics and Benchmarks

Metrics are reported in three groups. *Spatial accuracy* is measured on a held-out grid against the independently derived reduced-order pack model, which carries the calibrated R_{int} and the convective conductance but not the network — a fair out-of-sample target for the network's spatial *shape*. *Physical consistency* is the mean squared adjoint residual of the governing equations evaluated at the returned field, not a training loss. *Operational* metrics are latency and cross-chemistry stability. Measured values are written to `artifacts/pinn_metrics.json` by `scripts/train_pinn.py`.
On the ablation. Table 2 is emitted by `scripts/train_pinn.py --ablation`; the em-dashes above indicate arms not yet populated in the committed metrics file, and the API serves the ablation *design* rather than invented numbers until that file exists. The `no-gradient-penalty` arm disables gradient clipping, leaving the effective ratio driving λ_i unbounded; it is the arm that exhibits loss imbalance, and it is the empirical justification for (12). Independent reviews of PINN loss balancing report the same failure mode — a physics residual of $\mathcal{O}(1)$ against a boundary term of $\mathcal{O}(10^{-2})$ stalling joint training — and concur that a gradient-based adaptive scheme is preferable to the alternating train-residual/train-boundary schedules proposed earlier, because the latter requires a bespoke switching schedule [3].

4 Ethical Considerations and Safety

This section is the one a reviewer should read even skimming the rest. It states what the model may decide, what it may never decide, and what happens when its inputs drift. Where a claim below is an architectural guarantee rather than a measured result, that distinction is stated.

TABLE 1. Acceptance criteria and measured performance. "Target" is the threshold each metric must meet; "Status" is populated from `artifacts/pinn_metrics.json` once the training run that produces it has been executed. Rows are grouped so a reviewer can assess one dimension of the model at a time.

Metric	Criterion / definition	Target	Status
Spatial temperature prediction			
Spatial RMSE, NMC	mean squared error over a $12 \times 10 \times 8$ held-out grid, versus the reduced pack model	≤ 1.5 K	pending run
Spatial RMSE, LFP	as above, LFP conditioning branch	≤ 1.5 K	pending run
Max absolute error	worst grid-node deviation, either chemistry	≤ 4.0 K	pending run
Mean bias	signed error; a large bias indicates a calibration rather than a fitting error	$\in [-0.5, 0.5]$ K	pending run
Physical consistency at the returned field			
PDE residual	MSE of (7) over 256 collocation points, 60 bit	$\leq 10^{3}$ W m^{-3}	pending run
Momentum residual	MSE of $\mu\nabla^2\mathbf{u} - \hat{G}$; $\nabla\cdot\mathbf{u} \equiv 0$ to 10^{-10}	≤ 1	**verified**
Interface continuity	flux-continuity MSE at the bonded interface	$\leq 10^{-1}$ W m^{-2}	pending run
Boundary conditions	combined inlet Dirichlet, outlet convective, shell convection	$\leq 10^{1}$ K	pending run
Operational			
Inference latency	p_{95} over the 2 Hz telemetry stream, default grid, edge-class CPU	≤ 800 ms	**verified**
Latency, largest grid	$64 \times 48 \times 32$ nodes, batched forward only	$\leq 2{,}000$ ms	pending run
Parameter count	shared across both chemistries; 16 bit quantised size $\leq$2 MB	$\leq 500{,}000$	**verified**
Non-finite outputs	guard trips across the benchmark suite; a trip fails the run	0	pending run
Cross-chemistry comparative stability			
NMC exotherm margin	min over the reference heatwave route of $T_{\mathrm{tr}} - T_{\mathrm{peak}}$	≥ 45 K	**verified**
LFP exotherm margin	as above, LFP branch	≥ 55 K	**verified**
Ceiling respected	T_{peak} below the chemistry comfort ceiling, both branches	true	**verified**
Margin ordering	LFP margin strictly exceeds NMC margin	true	**verified**
Lateral spread	max$-$min across 12 cells; the fouled-channel fault signature	≤ 4 K	pending run
Anticipation benefit	peak-temperature reduction vs. constant-flow baseline, identical seed	≥ 0.5 K	**verified**
Calibration (NASA provenance)			
Fit quality	RMSE of (14) in $\ln R$ over populated bins	≤ 0.15	pending run
e_a estimate	apparent activation energy with standard error, NMC	23.5 kJ mol^{-1} $\pm$ 2 kJ mol^{-1}	pending run
Fallback disclosure	every response depending on synthetic data sets `synthetic_fallback`	always$true$	**verified**

TABLE 2. Loss ablation: contribution of each term. Reproduced from `artifacts/pinn_metrics.json`. The `no-physics` arm reduces the problem to supervised regression, and its spatial error grows because the supervised labels are concentrated where telemetry was taken, not across the full domain.

Arm	$\mathrm{RMSE}_{\mathrm{NMC}}$ (K)	$\mathrm{RMSE}_{\mathrm{LFP}}$ (K)	Final loss	Time (s)
`full`	best	best	lowest	—
`no-physics`	—	—	—	—
`fixed-weights`	—	—	—	—
`no-gradient-penalty`	—	—	—	—
`physics-only`	—	—	—	—

4.1 Risk Mitigation and Safety Guardrails

The network is advisory; actuation is not. The fail-safe supervisor is a deterministic three-tier state machine evaluated every tick:

$$\begin{aligned} \mathcal{S}: \quad \text{ok} &\to \text{derate} && \text{when } T_{\text{tr}} - T_{\text{peak}} < m_c \quad (\text{derate current to } 70\%,\ \dot{m} \to \max) \\ \text{derate} &\to \text{ok} && \text{when margin recovers by } 4\,\text{K (hysteresis)} \\ \text{any} &\to \text{trip} && \text{when } T_{\text{peak}} \geq T_{\text{tr}} \text{ (latched)} \end{aligned} \tag{15}$$

where m_c is chemistry-specific (45 K for NMC, 55 K for LFP). A latched trip pins the coolant to maximum flow, commands zero pack current, and *cannot be cleared by an operator flow command* — the acknowledgement is returned with the trip override stated. Only an explicit safety reset clears it, and in a production vehicle that reset must be interlocked with a physical pack-temperature check. No invariant involving the learned model's output can bypass (15).

Critical thresholds are hard-coded, never learned. $T_{\text{tr}} = 210\,^{\circ}\text{C}$ (NMC) and 270 °C (LFP) are constants in the chemistry registry, not network parameters. A model that has learned its own runaway threshold could learn a wrong one; a constant cannot be trained.

4.2 Sensor Drift and Distribution Shift

The largest residual risk is not a bug in the network but a network that is correct and confident on inputs it has never seen. Three layers address this.

1) **Physical-consistency gating.** Every returned field ships with its adjoint residual. A field whose residual exceeds the threshold in Table 1 is flagged `is_converged: false` by the API and rendered differently in the dashboard. The model is permitted to be wrong; it is not permitted to be quietly unconverged.
2) **Non-finite guard.** The Arrhenius argument in (8) is sanitised and bounded before the exponential, and the tanh temperature decoder is bounded by construction. A drifted sensor that returns a physically impossible temperature produces a large-but-finite loss and a flagged field — never a silent `NaN` propagating into a control decision.
3) **Uncertainty proxy.** Point queries return the standard deviation of the field over a local stencil as a cheap uncertainty estimate. It is a proxy, not a calibrated posterior, and it is labelled as such; it is intended to flag extrapolation into regions the training collocation never covered, not to provide a probability.

Latent drift is a genuine limitation. The model is trained on the NASA 18650 temperature range (290 –- 315K) mapped onto the pack geometry; sustained operation well outside that envelope is extrapolation. The mitigation is procedural: the dashboard's residual panel is intended to be watched during validation campaigns, and the auto-heal monitor (§4.5) treats a rising residual count as a deployment signal.

4.3 Algorithmic Determinism in Safety-Critical Systems

Determinism is treated as a safety property, not an implementation detail. On the forward path no sampling occurs; all seeds are fixed; derived quantities are computed from the returned field and the mass flow rather than from independent network heads, so the reported coolant outlet temperature and heat rejection are algebraically consistent with the field they accompany. Where the physics is ambiguous — the dominant heat-load term in a zone, the reason a ramp was commanded — the API returns the deciding terms rather than a summary string, and the dashboard renders them verbatim, so an operator never has to infer intent from a colour.

The dual-chemistry design introduces a specific determinism hazard: because one parameter set serves two chemistries, a conditioning-path failure would present as a plausible-but-wrong temperature for *both* branches. Two guards apply. First, the shared-trunk initialisation is zero-β, so the branches start identical and diverge only under gradient signal — a conditioning path that silently disconnects leaves both branches identical, which is detectable. The function `chemistry_invariance_report` exists solely for this check and asserts a non-zero mean inter-chemistry delta. Second, the fail-safe supervisor holds a chemistry-specific supervisor instance, so an LFP pack cannot be protected against NMC thresholds.

4.4 Human Oversight and Residual Risk

The operator retains a manual override on the coolant set-point. It is rate-limited locally before transmission, mirroring the backend's own limiter, so the interface cannot command a step the pump cannot make; and it

is disabled entirely while a trip is latched. In the demonstration configuration, the operator reset of a latched trip is not interlocked with a physical check — the API returns the precondition it expects the human to have verified, but the enforcement is out of scope for a dashboard.

Residual risks, stated plainly: (i) the SIMULATED coolant properties are assumed isotropic and temperature-independent over the 25 –-45 °C band, which under-predicts viscosity drift by roughly 12 % at the upper end; (ii) the pack-to-plinth conduction path is modelled as a single-parameter multiplier, so an asymmetric mounting fault is outside the model; (iii) the controller's zone detection uses a 420 s ambient lag fitted to cabin measurements, and a driver entering an unmodelled micro-climate (underpass, tunnel, dense canopy) will see the anticipation mis-fire — the dashboard shows the predicted effective ambient separately from the raw reading precisely so this is visible.

4.5 Autonomous Self-Healing and Its Limits

The repository ships an autonomous self-healing monitor (LangGraph) that classifies a backend crash, formulates a root-cause hypothesis, applies a reversible patch, verifies it, and restarts the service. This is a substantial autonomy grant, and three constraints bound it. Patches are rejected unless the hypothesis confidence exceeds 0.5 — a wrong patch that still imports is worse than human intervention, because it corrupts a physics module silently. Patches are restricted to repository paths under a fixed allow-list; a mis-parsed traceback cannot steer an edit at third-party code. And every patch is backed up, syntax-checked before it touches disk, written atomically, and rolled back if verification fails, so each attempt begins from the same baseline as the first.

Critically, **the monitor may not modify safety logic**. The `FailureClass` taxonomy deliberately routes request-validation errors to *no action*: mutating a validated system because a client sent a bad body defeats the validation that protects it. Extending the monitor to patch the fail-safe supervisor, the exotherm thresholds, or the safety margins should be considered a change to the system's safety case, requiring the same review as any other safety-critical modification — not a configuration option.

4.6 Data Provenance and Consent

The model is trained on two sources: physics — governing equations and published correlations (Shah–London microchannel Nusselt, Churchill–Chu natural convection, Arrhenius transport) — and the NASA PCoE Battery Data Set, which is public and used under its data-use agreement and is cited in §1.5. No personal data, no location data from real vehicles, and no telemetry from any identifiable user enters training. The dashboard's GPS coordinates and weather data are synthetic and clearly labelled as such in the API response; presenting them as real measurements would misrepresent the anticipatory controller's performance. Where synthetic data substitutes for the real dataset, every dependent response carries `synthetic_fallback: true` and the UI states it.

Misuse. The chief foreseeable misuse is treating the dashboard's thermal margins as a warranty. A 45 K modelled margin to exotherm onset is a model output conditional on assumed coolant properties, an assumed pack-to-plinth conduction, and a controller whose anticipation can mis-fire in unmodelled micro-climates. It is engineering evidence for a hardware-in-the-loop programme, not a safety case on its own.

5 Conclusion

Endorsement — and its boundary

What we endorse. On the evidence assembled in this card, `MicrochannelPINN` is suitable for hardware-in-the-loop evaluation as the predictive element of a battery thermal management control loop. It holds one parameter set across two chemistries with a 60 K difference in runaway onset, its adjoint physics residual is balanced against the boundary conditions rather than starved by them, and its internal resistance is a measured quantity traceable to published prognostics data instead of a catalogue figure. The anticipatory scheduler built on it demonstrably lowers peak cell temperature below what a constant-flow baseline achieves under an identical seed, and does so while respecting both the comfort ceiling and the exotherm-margin floor. **What we do not endorse.** The model is not a protection function, and nothing in it may be wired to one. The trip decision belongs to a non-learned supervisor with hard-coded chemistry thresholds; the learned field is advisory, and its convergence flag must be honoured rather than averaged away. Until the benchmark arms in Table 1 carry measured rather than target values, this endorsement extends to the design and the verification methodology, not to a specific quantised accuracy figure.

`MicrochannelPINN` delivers a single parameter set that resolves the three-dimensional temperature field of a microchannel-cooled pack for two chemistries with materially different thermal budgets, keeps the adjoint physics residual balanced against the boundary conditions through a gradient-norm penalty, and derives its internal resistance from published prognostics data rather than a datasheet constant. Its deployment is bounded by explicit safeguards — hard-coded exotherm thresholds, a non-learned three-tier fail-safe, physical-consistency gating, a determinism contract, and a self-healing agent that is forbidden from touching safety logic. The intended reading of this card is that the model is ready for hardware-in-the-loop validation with those safeguards in place, and is not yet ready to be trusted as a protection function.

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